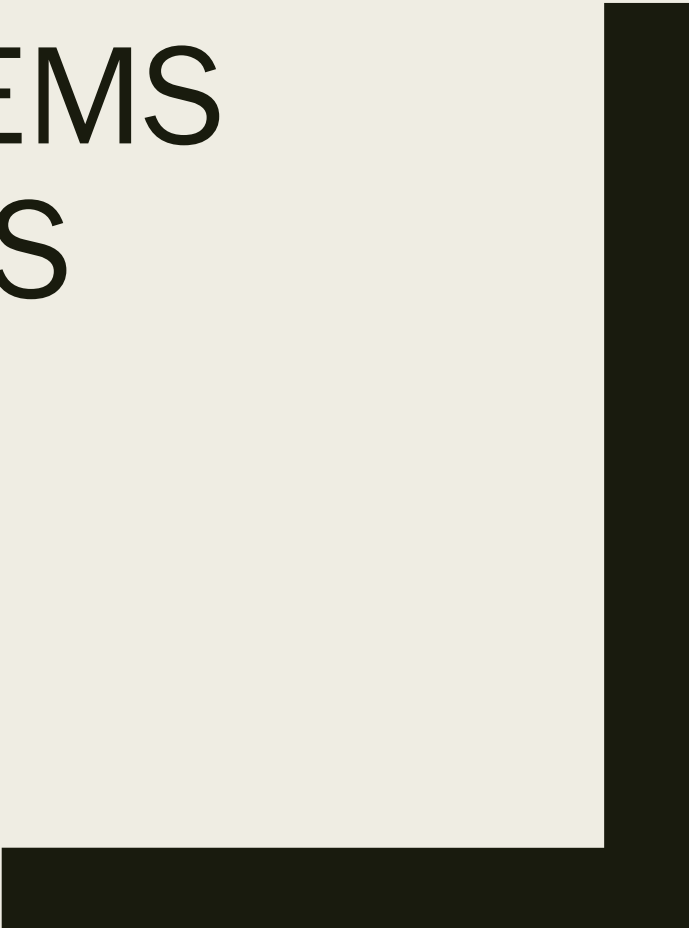




COMPUTER SYSTEMS FUNDAMENTALS (4COSC004W)

Week 1. Part 2 of 2



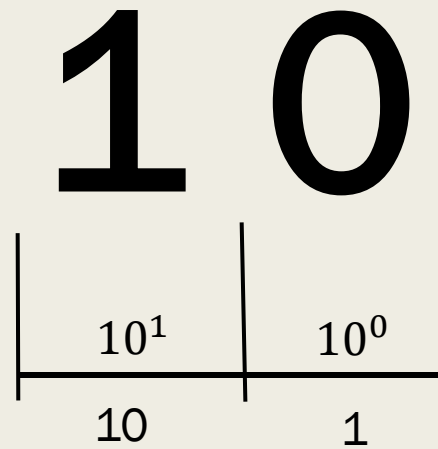
BINARY – BASE 2

Positive Integers

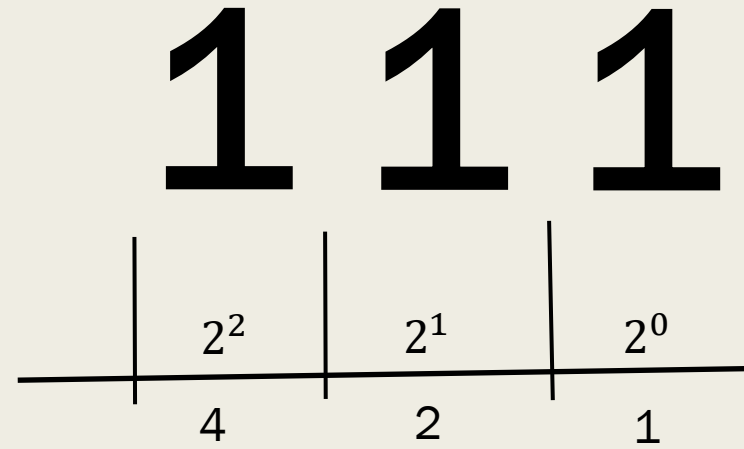
By the end of this lecture, you will:

- Be able to convert Binary Positive Integers to their Denary values
- Be able to convert Denary Positive Integers to their Binary values
 - *Division by 2 & observation techniques*
- Quick parity check
- The Bit
- The Nibble

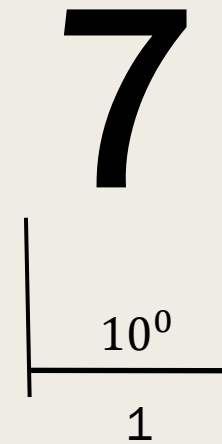
Decimal / Denary – Base 10



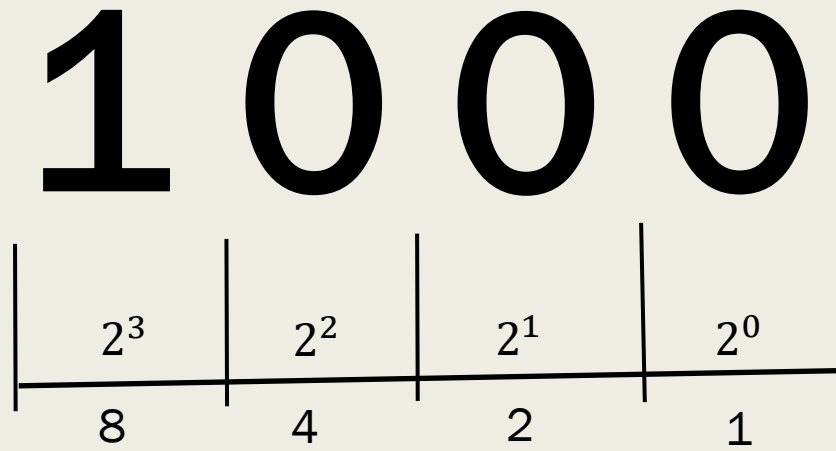
Base 2
Binary



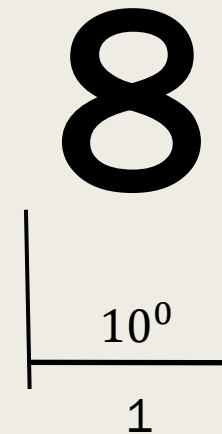
Base 10
Denary



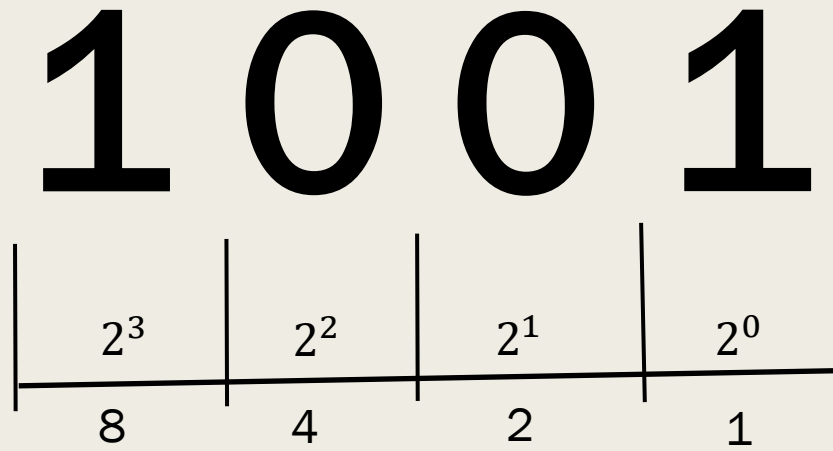
Base 2
Binary



Base 10
Denary



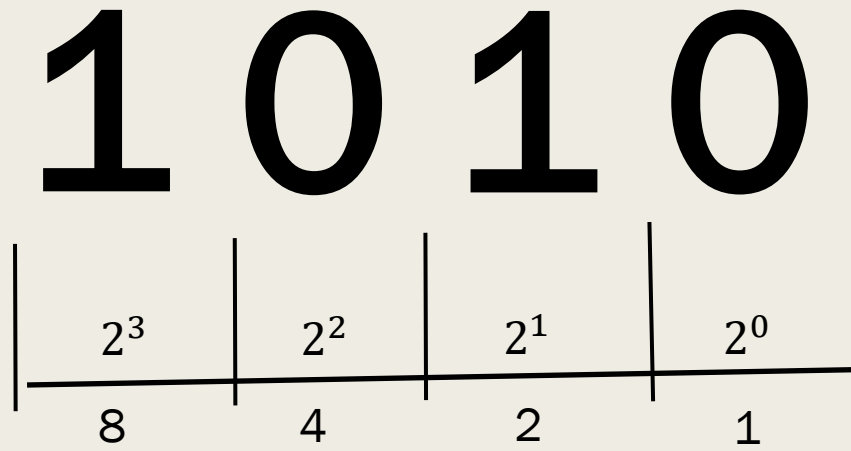
Base 2
Binary



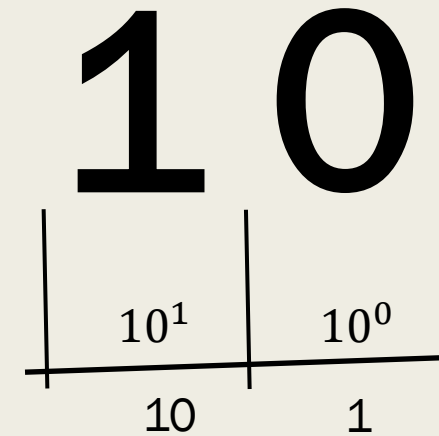
Base 10
Denary



Base 2
Binary



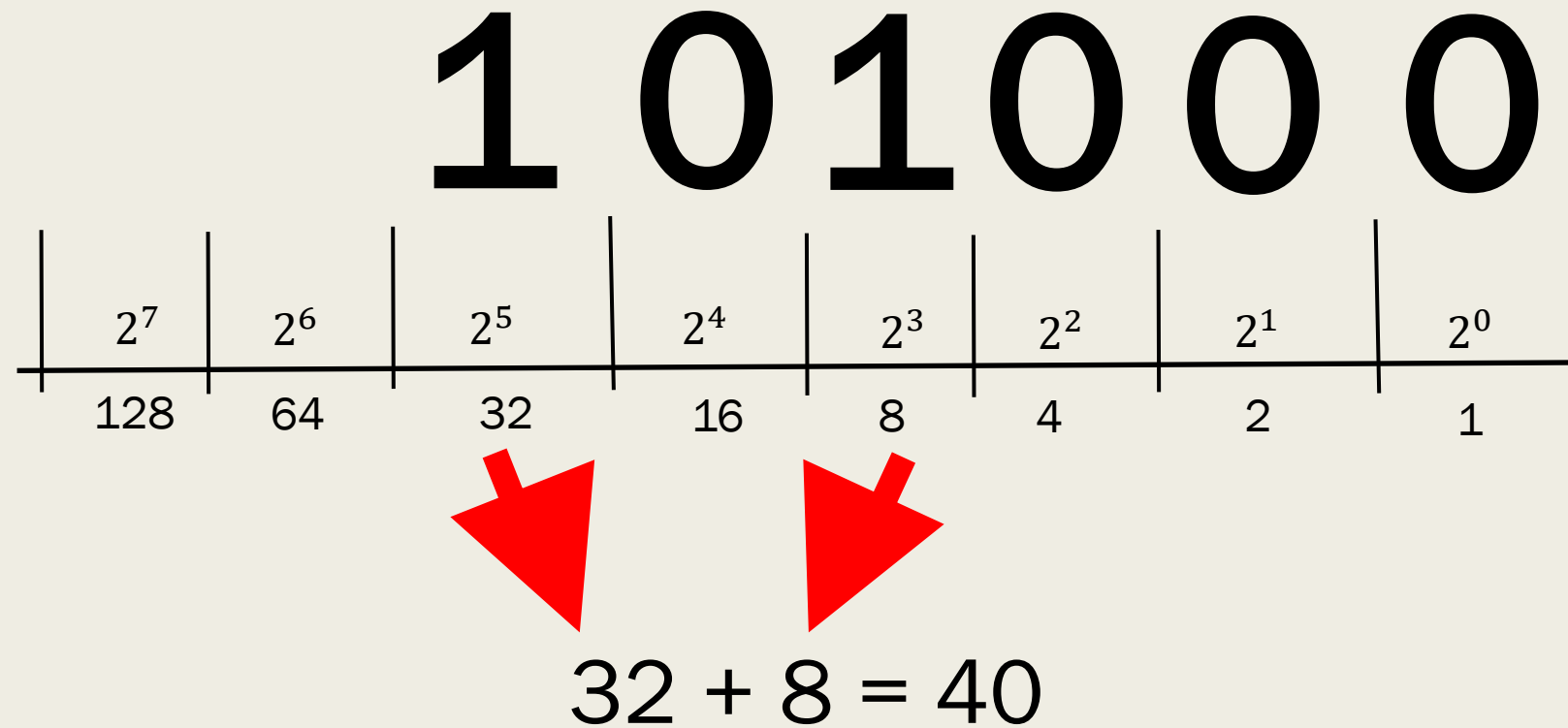
Base 10
Denary



Hexadecimal table:

Denary	Binary				Hexadecimal
0	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	2
3	0	0	1	1	3
4	0	1	0	0	4
5	0	1	0	1	5
6	0	1	1	0	6
7	0	1	1	1	7
8	1	0	0	0	8
9	1	0	0	1	9
10	1	0	1	0	A
11	1	0	1	1	B
12	1	1	0	0	C
13	1	1	0	1	D
14	1	1	1	0	E
15	1	1	1	1	F

Converting Binary to Denary



Converting Binary to Denary

1	1	0	1	1	1	1	1
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

$$128 + 64 + 16 + 8 + 4 + 2 + 1 = 223$$

Converting Denary to Binary

40 / 2 = 20 remainder 0

20 / 2 = 10 remainder 0

10 / 2 = 5 remainder 0

5 / 2 = 2 remainder 1

2 / 2 = 1 remainder 0

1 / 2 = 0 remainder 1



1 0 1 0 0 0

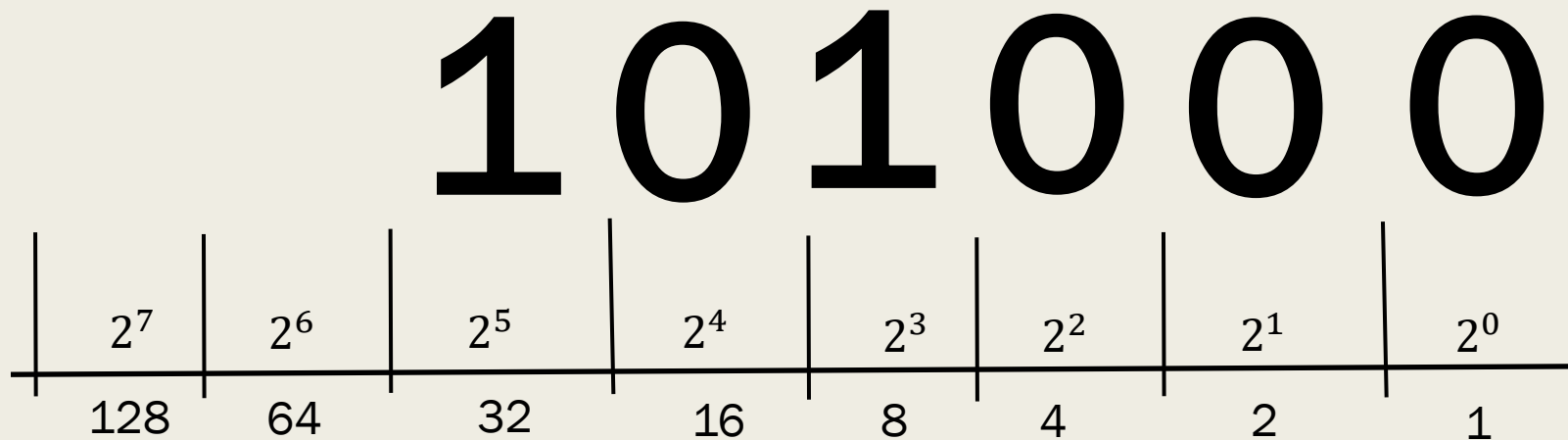
Converting Denary to Binary

Denary 40:

Which is the largest power of 2 which fits in 40?

Now subtract the 32 from the original 40

Which is the largest power of 2 which fits in 8?



Converting Denary to Binary

43 / 2 = 21 remainder 1
21 / 2 = 10 remainder 1
10 / 2 = 5 remainder 0
5 / 2 = 2 remainder 1
2 / 2 = 1 remainder 0
1 / 2 = 0 remainder 1



1 0 1 0 1 1

Converting Denary to Binary

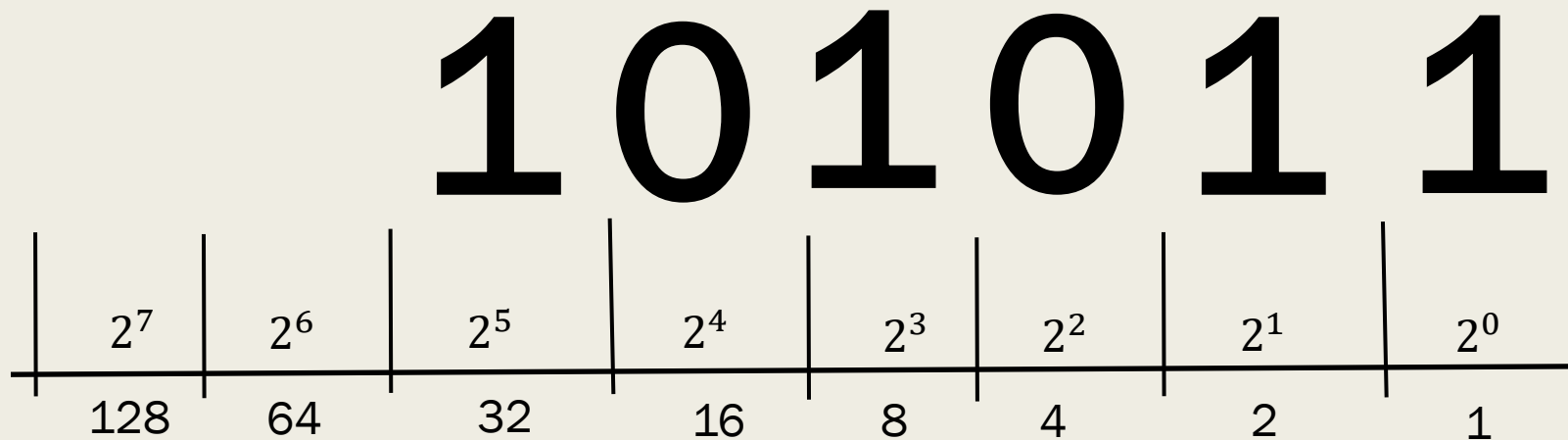
Denary 43:

Which is the largest power of 2 which fits in 43?

Now subtract the 32 from the original 43

Which is the largest power of 2 which fits in 11?

Now subtract the 8 from the 11



The Byte, The Nibble & The Bit

- Byte:
 - *An 8-Bit Binary Value*
 - 256 possible values
 - 2^8
- Nibble:
 - *A 4-Bit Binary Value*
 - 16 possible values (2^4)
- Bit:
 - *Single Binary Bit*
 - 0 or 1
 - 2^1

Parity check

- If the Denary value is **odd**
 - *The **Least Significant Binary Bit** will be **1***
- If the Denary value is **even**
 - *The **Least Significant Binary Bit** will be **0***

Try it out for yourself - task 1

- Write down an 8-Bit binary value
- Try to convert it into Denary
 - *Start with the Least Significant Digit (on the right)*
 - *Work through all the Bits*
 - *Add up the Denary values of each Bit*
- Write down the Denary value of the Binary value you started with
 - *On another piece of paper*
- Try another

Try it out for yourself - task 2

- Try to convert each of the Denary values,
 - *back to their Binary representation*
 - *Try both methods*
 - *Make sure they both give you the same result*

An exercise for you:

Convert the **Denary** number 123 into **Binary**:

Converting Denary to Binary

123 / 2 = 61 remainder 1

61 / 2 = 30 remainder 1

30 / 2 = 15 remainder 0

15 / 2 = 7 remainder 1

7 / 2 = 3 remainder 1

3 / 2 = 1 remainder 1

1 / 2 = 0 remainder 1



1 1 1 1 0 1 1

Converting Denary to Binary

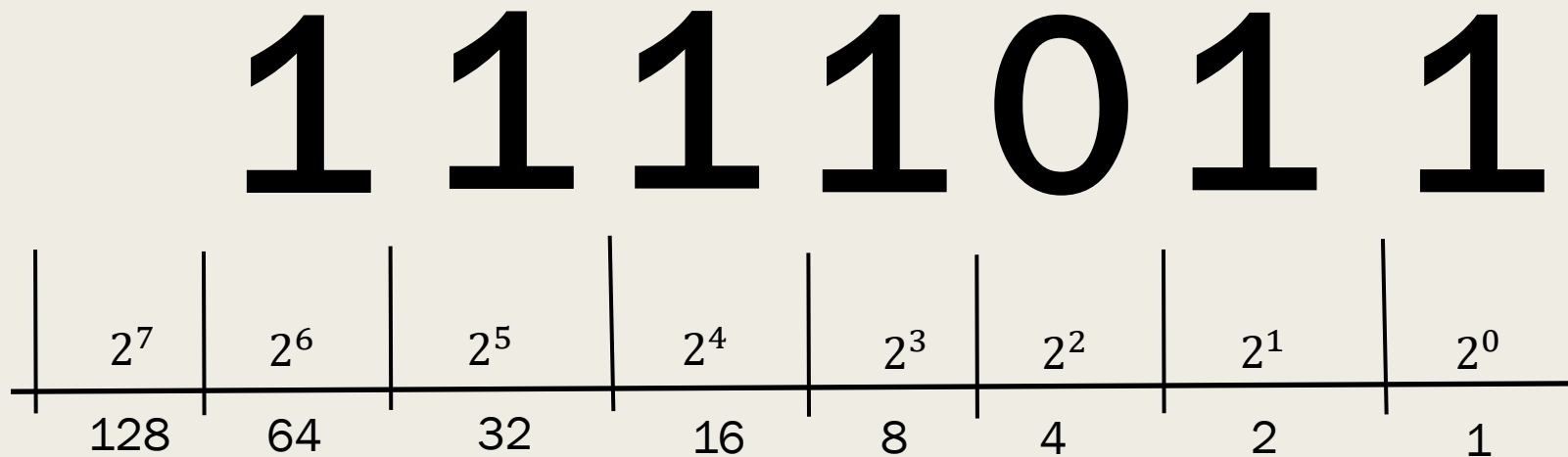
Denary 123:

Which is the largest power of 2 which fits in 123?

Now subtract the 64 from the original 123

Which is the largest power of 2 which fits in 59?

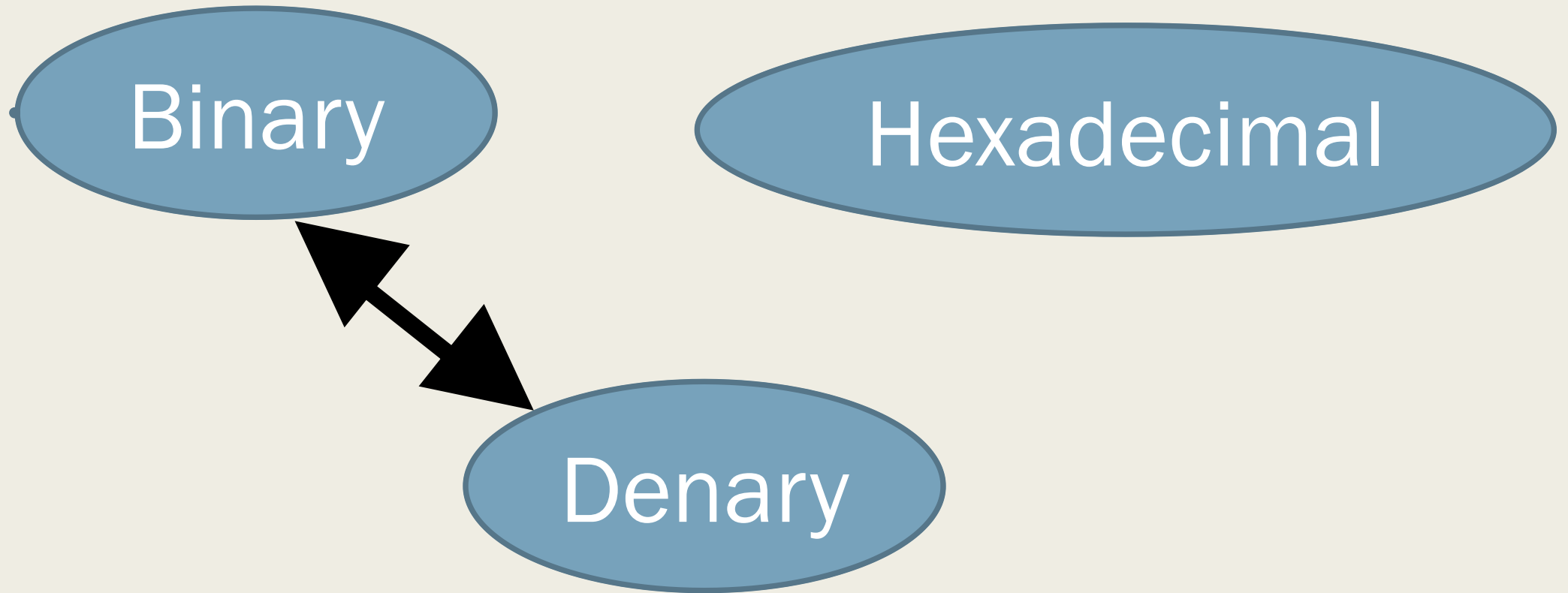
Now subtract the 32 from the 59



In this part we have covered:

- Binary:
 - *Converting from Binary to Denary*
 - *Converting from Denary to Binary*
 - Division by 2 method
 - Observation method

Number System Triangle



Tutorial exercise:

- This will provide you with:
 - *Binary values to convert to Denary*
 - *Denary values to convert to Binary*
- By the end of this week's tutorial you will be familiar with the *language* computers use for data
 - *To store and read*

In the next lecture we will cover:

- Hexadecimal:
 - *Theory*
 - *Denary – Binary – Hexadecimal triangle*
 - *Why use Hexadecimal*
 - *Quick Binary - Hexadecimal conversions*
- Binary operations
 - *Logical*
 - *Addition*

Thank you

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These slides have been edited reviewed and amended by Adem Coskun, Izzet Kale and George Charalambous.
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